

Ishaan Jain

Semiconductor Engineer

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SUMMARY

Semiconductor engineer with an MSc in NanoScience from KU Leuven (Cum Laude) and a BTech in Engineering Physics from IIT Delhi. Spent 14 months at IMEC modeling middle-of-line parasitics for CFETs in Synopsys TCAD, with Python tools that hold capacitance error under 10%. Has taped out and measured an OTA in 180 nm and built a 400 V battery pack with custom PCBs. Seeks a device, process or equipment engineering role.

EDUCATION

MSc, NanoScience, Nanotechnology and Nanoengineering, KU Leuven, Leuven, Belgium Sep 2024 – Aug 2026
Specialisation: Nanodevices and Circuits | 70% (KU Leuven scale, Cum Laude)

Thesis (at IMEC): Device Technology Co-Optimization (DTCO) by modeling MOL parasitics for advanced CFETs

BTech, Engineering Physics, Indian Institute of Technology Delhi, New Delhi, India Nov 2020 – May 2024
Minor in Electrical Engineering | First Class, 75%

Thesis: Design and Simulation of Single Photon Avalanche Diodes (SPADs) in Silvaco Atlas TCAD; 30% PDE at 800 nm, breakdown voltage -18.1 V

WORK EXPERIENCE

DTCO Research Intern → MSc Thesis Researcher Jul 2025 – Aug 2026
IMEC, Leuven, Belgium

- Performed DTCO by simulating 3D CFET structures in Synopsys TCAD (sdevice, svisual, Raphael FX) and extracting SPICE models with parasitics for logic standard cells and SRAM bitcells.
- Parametrised the CFET structure across 43 geometric and design parameters and wrote sweep scripts, including multi-parameter combined sweeps, for design-space exploration.
- Wrote Python automation, used by others on the team, that reads svisual log files, computes capacitances from device dimensions and generates SPICE netlists.
- Reduced the middle-of-line parasitic network to input/output/node capacitances, holding individual values under 10% error, by applying Kron reduction.

Intern, Opto-Mechanical Instrument Engineer Jun 2023 – Aug 2023
Tata Institute of Fundamental Research, Mumbai, India

- Designed a robotic pick-up arm for a telescope spectrograph to 16 μm positional and 5 μrad angular precision, using SolidWorks, Ansys FEA (structural and thermal) and Zemax for verification.
- Allocated the error budget across vibration, thermal effects and tolerance stack-up, and ran FMEA and DFMA so the parts could be machined and assembled.
- Assembled and fine-tuned the prototype, including microscope-assisted micro-assembly; worked the design through with the institute's mechanical, electrical, optics and manufacturing heads.

Robotics and IoT Content Developer Jun 2022 – Jul 2022
Rancho Labs, Delhi, India

- Designed and taught an introductory robotics course (electronics, Arduino C++) to ~75 school students, and was their first point of contact when builds stopped working.

PROJECTS

RISC-V CPU Core and Low-Power Design, Course projects, KU Leuven 2024 – 2026

- Wrote a 5-stage pipelined RISC-V core in Verilog, checked with a self-checking SystemVerilog testbench, and ran a Python-scripted synthesis and place-and-route flow to compare power, speed and area.

Optical NEMS-Based XOR Gate, Course project, IMEC Feb 2025 – Jun 2025

- Reached 0.8 μm plate displacement under a 10 V difference to modulate optical transmission at 1550 nm by optimising a published NEMS structure in COMSOL and MATLAB.

Telescopic-Cascode OTA, Course project, MICAS, KU Leuven Sep 2024 – Nov 2024

- Designed and taped out an OTA in 180 nm (Cadence Virtuoso, ADE) after PVT corner and Monte Carlo mismatch simulations; measured 87.7 dB gain, 1.23 MHz GBW and 1 mW at 1.8 V.
- Characterized the silicon across supply-voltage and temperature sweeps; results matched simulation within 1%.

Total Solar Eclipse Expedition, Equipment Lead, Exmouth, Australia

Dec 2022 – Apr 2023

- Designed telescope mounts and filters (3D printing, laser cutting), load-tested them before the event and captured the solar corona on 4 telescopes at 30 fps per camera.
- Specified and sourced all hardware, tracked the INR 3 lakh hardware budget, shipped the gear with customs paperwork and repaired tools on site.

XLR-20 Electric Racecar, Formula Student (AXLR8R), IIT Delhi

May 2021 – Jun 2022

- Co-led a ~INR 10 million build and designed a 400 V lithium-ion battery module with a custom passive BMS; 3rd overall at Formula Bharat 2022.
- Designed two PCBs in Altium to monitor all 192 cells, debugged 20+ PCBs, and ran 50+ hours of testing on the cells and boards with no electrical failures.

SKILLS

Semiconductor and EDA: Synopsys TCAD (sdevice, svisual, Raphael FX, Workbench), Silvaco Atlas, Cadence Virtuoso, ADE, Genus, Innovus, Xcelium, LTSpice, COMSOL Multiphysics

Lab and characterization: Probe station, C-V and I-V measurement, source-measure units, LCR meters, oscilloscopes, AFM, ellipsometer, profilometer, sputtering, dry etch (RIE), CVD, wafer handling, ESD-safe practice

Programming and design: Python (NumPy, Pandas), MATLAB/Simulink, Verilog, SystemVerilog, UVM, C/C++, Linux shell, Git; Altium, SolidWorks, Ansys Workbench (FEA), Zemax, STM32

Methods: FMEA, 5-whys root-cause analysis, DFM/DFMA, error-budget analysis, tolerance stack-up, control charts, board bring-up and debug, HV safety practice

RELEVANT COURSEWORK

Semiconductor Devices; Technology of Integrated Systems (semiconductor processing, lithography); Reliability and Yield; Physical Material Characterization; Digital IC Design; Mixed Signals Design and Implementation

PUBLICATIONS

Shah, H., Ninan, J. P., Poojary, S. S., ... Jain, I., ... et al. (2024). Mechanical design of the pick-up arm prototype for TIFR-ARIES Multi-Object Optical to Near Infrared Spectrometer (TA-MOONS). Proc. SPIE 13096, 130968P. <https://doi.org/10.1117/12.3019499>

HONORS AND AWARDS

- National STEM honors: Olympiad OCSC selectee in Physics and Astronomy, HBCSE (national training camp for India's International Olympiad teams); JEE Advanced AIR 1756; KVPY Fellow (Govt. of India DST); Sakura Science Exchange Fellow (Japan, 2024).
- Certification: Cleanroom Fundamentals & Semiconductor Technologies, University at Buffalo (Coursera), 2026.

ADDITIONAL DETAILS

Work authorisation: Belgium (no sponsorship required under search year visa); rest of EU: EU Blue Card eligible

Availability: Available immediately; open to relocating; open to 24/7 shift work (nights, rotating shifts, on-call)

Languages: Hindi (native), English (fluent, IELTS 7.5), Dutch (A2)

Leadership: Executive Head, Physics and Astronomy Society, IIT Delhi (2021–2023); IEEE Student Member